

## **REQUEST FOR INFORMATION (RFI) 70RWMD26RFI00000003**

### **Title: Novel Surface Enhanced Raman Spectroscopy (SERS) Scanning Technology for Biodetection**

**NOTE: THIS IS A REQUEST FOR INFORMATION (RFI) ONLY.** This RFI is issued solely for information and market research planning purposes. This RFI shall not be construed as an obligation on the part of the Government. This is NOT a Request for Quotations or Proposals. No solicitation document exists, and the Government may or may not issue a formal solicitation as a result of the responses received to this RFI.

The Government will not pay for any response or demonstration expenses. All costs incurred responding to this RFI will be solely at the interested party's expense. Failure to respond to this RFI will not preclude participation in any future solicitation. Any information received will become the property of the Government and will not be returned to the submitter. Interested parties are responsible for adequately marking proprietary or sensitive information.

The Government's explicit intent, in this request for information, is to not receive from respondents any proprietary data, trade secrets, business sensitive information, or information considered CONFIDENTIAL under 18 U.S.C. §1905. The Government's constraint does not in any way relieve contractors from their responsibility to properly mark proprietary data when it is provided, nor does it alleviate any requirement for the Government to protect marked data. The Government is not obligated to protect unmarked data. The information provided in this RFI is subject to change and is not binding on the Government. All submissions become the property of the Federal Government and will not be returned.

### **DESCRIPTION**

It is imperative to rapidly detect and identify biothreats caused by a mass casualty incident or a lone wolf attack. Rapid detection and accurate identification are critical in the response effort led by civil support teams and first responders thus mitigating the impact of biothreats. Surface-enhanced Raman spectroscopy or surface-enhanced Raman scattering (SERS) is a surface-sensitive technique that enhances Raman scattering by molecules absorbed on rough metal surfaces or by nanostructures that are used to detect the presence of very low concentrations of analytes involving structural and molecular identification. SERS has high specificity and sensitivity of surface signal due to surface enhancement mechanisms occurring at the surface. These surface enhancements are due to electromagnetic fields of the metallic enhancer and the surface components of the chemical enhancer. For bio the chemical enhancer can be structural components on the surface of cell. Microorganisms produce unique biosignatures that are read as spectra at different wavelengths when analyzed with spectrometry applications. SERS technology amplifies the signal that produces the biosignature with the use of metallic nanoparticles. Nanoparticles can be silver, gold, or another metallic and are coupled with the chemical enhancer. This allows for samples with low concentrations to be detected.

SERS is widely used in the materials chemistry community and the increasing exploration for biodetection use cases due to its high sensitivity and specificity. Currently, there are no SERS biodetection capabilities used to detect biothreats such as select agents and is not widespread in routine biodetection processes. Challenges such as reproducibility, standardization, and the development of portable devices need to be fully addressed to facilitate broader application. CWMD is committed to protecting the American people from biological threats by understanding technological advancements and collaborating with partners across the homeland security enterprise.

## **PURPOSE OF THIS REQUEST FOR INFORMATION (RFI)**

The CWMD Office seeks to explore the utility of surface enhanced Raman spectrometry in biodetection and to develop a SERS nanostructure prototype to be used to detect bioagents of interest to DHS components and their stakeholders.

## **INFORMATION REQUEST**

The DHS CWMD office is interested in exploring the potential application of SERS scanning technology for the detection of bioagents, specifically select agents. Research and development efforts would focus on evaluating the effectiveness of SERS scanning technology as a tool for biodetection. The Raman application associated with SERS technology may utilize either a commercially available off-the-shelf (COTS) device or a newly constructed device. If a COTS device is proposed, a plan addressing potential interoperability issues should be included. Additionally, the technology would be paired with an adaptive signature library to support sample identification.

The anticipated requirement may involve a period of performance of up to 36 months. Offerors should provide a detailed approach plan that justifies the proposed length for the period of performance. Key components that CWMD seeks to assess through this project include the following:

### **Detailed Requirements:**

1. SERS nanostructure prototype must adhere to cell surface proteins of a select agent
2. SERS nanostructure must be designed to give the optimal spectral signal for select agents
3. Raman platform works efficiently with SERS nanostructure prototype
4. Adaptive library must be cloud based
5. Adaptive library should have the capability to update monthly

## **RESPONSES TO THE RFI**

Interested parties should submit electronic responses to the point of contact identified below:  
Ariana Thompson, Contract Specialist, [Ariana.Thompson@hq.dhs.gov](mailto:Ariana.Thompson@hq.dhs.gov).

RFI responses should be in .docx format. Figure legends, labels, and table of contents may be no smaller than font size 8 as presented in Microsoft Word. Responses shall be limited to 10 pages inclusive of cover page containing respondent information. Proprietary information, if any, should be minimized and **MUST BE CLEARLY MARKED**. The subject line of the email should contain the RFI number (70RWMD26RFI00000003) and "Novel Surface Enhanced Raman Spectroscopy (SERS) Scanning Technology for Biodetection."

At a minimum, responses should include:

1. Company information: (name, phone number, UEI, CAGE Code, POC information, website, etc.).
2. GSA contract(s): contract numbers and contractor's socioeconomic status.
3. Description of your company's capabilities as they relate to the services described in this notice.

4. A description of your company's previous experience providing the services described in this notice, including independent testing that substantiates your capabilities.
5. A description of any teaming/subcontracting your company anticipates utilizing in support of this requirement.
6. A statement indicating the ability of your company to meet the proposed period of performance (POP) OR a detailed explanation of what a feasible POP to complete this effort would look like.
7. A proposed method, timeline and cost estimate to complete the above-described research and development effort.
8. Any other relevant information that is not listed above which the Government should consider in finalizing its market research and acquisition strategy.

## **Glossary A: Terms and Definitions**

**Raman spectroscopy-** Raman spectroscopy is a technique that involves the inelastic scattering of light to analyze molecular vibrations, providing valuable information about the structure and composition of molecules.<sup>1</sup>

**Surface-enhanced Raman spectroscopy (SERS)-** Surface-enhanced Raman spectroscopy (SERS) is defined as a sensitive method that enhances the Raman scattering of molecules using nanostructure materials, such as plasmonic metals, allowing for the detection of single molecules through plasmon-mediated amplification of electrochemical fields or chemical enhancement.<sup>2</sup>

**Nanostructure-** Nanostructures are defined as nanoscale objects that behave as whole units in terms of their transport and properties, typically ranging from 0.2 to 100 nanometers in dimension. They include nanoparticles, nanoclusters, and nanocrystals, with nanoclusters having at least one dimension between 1 and 10 nanometers.

**Electromagnetic field-** a field of force that consists of both electric and magnetic components, resulting from the motion of an electric charge and containing a definite amount of electromagnetic energy.<sup>3,4</sup>

**Select agents-** select agents refer to biological agents and toxins determined by the Federal Government to pose a severe threat to public health, safety, and animal or plant health and products. These agents are subject to regulatory oversight to ensure their safe use in research while protecting public health.<sup>5</sup>

## **References**

1. [Raman Spectroscopy - an overview | ScienceDirect Topics](#)
2. [Surface Enhanced Raman - an overview | ScienceDirect Topics](#)
3. [Nanostructure - an overview | ScienceDirect Topics](#)
4. [ELECTROMAGNETIC FIELD Definition & Meaning | Dictionary.com](#)
5. [Select Agent - an overview | ScienceDirect Topics](#)